

Eshwar Pamula

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EDUCATION

The Ohio State University

Bachelor of Science in Mechanical Engineering with Honors Research Distinction

Minor: Economics

Columbus, OH

Graduation: December 2025

GPA: 3.36/4.0

PROFESSIONAL EXPERIENCE

Flux Marine

Powertrain Engineering Intern

Bristol, RI

May 2025 - August 2025

- Developed an assembly fixture with a custom quick-release clamp mechanism, optimizing alignment precision and ensuring tight wrapping of all segmented stator core pieces for consistent motor assembly.
- Designed and built a custom shrink-fit jig using SolidWorks modeling, 3D printing, and CNC machining, reducing assembly process time by 10 seconds while ensuring flush seating of all motor housing segments.
- Conducted comprehensive dynamometer testing protocols across varying torque and RPM ranges up to 4000 RPM using Grafana data analysis platform, validating motor performance metrics for future design iterations.
- Contributed to cross-functional product development across 3 different motor builds at a Series-A marine propulsion startup.

Allegion Inc.

R&D Mechanical Engineering Intern

Colorado Springs, CO

May 2024 - August 2024

- Worked on new product development initiatives for two new electro-mechanical residential lock set designs.
- Designed a dual-CAM interconnect assembly using CREO and validated it through FEA simulations for BHMA standards.
- Manufactured early prototypes (3D printing and CNC) of the lock system, ensuring feasibility for the 2025 product pipeline.

Ecosystem Energy

Mechanical Engineering Intern

Columbus, OH

May 2023 - August 2023

- Monitored and analyzed daily control data from heat exchangers to enhance energy efficiency and system optimization.
- Led a \$300K project for energy control using pneumatic valves and achieved a 27% reduction in energy consumption.

Engie North America

Project Engineering Intern

Columbus, OH

October 2022 - May 2023

- Conducted comprehensive engineering reviews of HVAC designs for Ohio State's Combined Heat and Power plant.

RESEARCH EXPERIENCE

Multi-functional Materials and Intelligent Design Lab

Undergraduate Researcher

Columbus, OH

May 2024 - Present

- Awarded College of Engineering (Honors Distinction) grant for AI-driven geometric primitive-based topology optimization.
- Developed algorithms to extract meaningful structural patterns from optimized structural designs, reducing solving time.

Gear and Power Transmission Laboratory

Undergraduate Research Assistant

Columbus, OH

April 2023 - May 2024

- Conducted experimental analysis of polymer (Torlon) gear properties using Symbrium 3-shaft-bending machines.
- Monitored and controlled oil flow, temperature, and torque conditions to assess thermal performance of gear systems.

LEADERSHIP AND INVOLVEMENT

Formula Buckeyes FSAE

Suspension Engineer

Columbus, OH

2022 - August 2024

- Designed and manufactured uprights, mounts, and hard points using the mill and lathe for the control arms and sway bar.
- Helped manufacture in-house Billet 7075-T6 Uprights using Kia SKT25 CNC to reduce stiffness and complexity.
- Contributed to designing and manufacturing a rear Anti-Roll Bar (T-Bar setup) with SolidWorks and Huron Lathe CNC.
- Played a key role in the team's success, securing a 3rd-place finish in 2023 and a 1st-place finish in 2024 at Michigan FSAE.

Ohio State University: College of Mechanical and Aerospace Engineering

Undergraduate Teaching Assistant

Columbus, OH

April 2023 - Present

- Taught and mentored 500+ students over five semesters in Statics, Mechanics of Materials, and Machine Elements II.
- Runner-Up for the Best MAE Undergraduate Teaching Assistant in 2025.

AWARDS, SKILLS, AND INTERESTS

- **Awards:** ShowOHIO 3rd place ('23), Ohio State Fintech Pitch Competition Winner ('23), OSHA 10 Certification
- **CAD & FEA:** SolidWorks, CATIA, CREO, AutoCAD, ANSYS
- **Skills & Tools:** CNC Machining, Finite Element Analysis (FEA), 3D Printing, Lathe, Mill, GD&T, Grafana
- **Programming Languages:** MATLAB, Python, Arduino IDE